

## Department of Computer Science and Engineering

### Individual Contribution Report — Team Assignment

*To be completed by EACH student individually and submitted along with the team's assignment. This report explains what YOU specifically did, so it must be written in your own words and reflect only your own work.*

#### A. Assignment & Team Information

|                          |                                                                                 |
|--------------------------|---------------------------------------------------------------------------------|
| Course Code & Name       | CSA0614 – Design and Analysis of Algorithms                                     |
| Assignment Title         | Conflict-Free Exam Timetable Scheduling Using Backtracking (Graph Colouring)    |
| Team / Group No.         | 3                                                                               |
| Team Members (all names) | Haresh Kumar N L (192425009), Nirmal Kasi (192424248), Mohan Kumar J (92424060) |
| Date of Submission       | 02/09/2026                                                                      |

#### B. Student's Own Details

|                     |                                       |
|---------------------|---------------------------------------|
| Student Name        | Mohan Kumar J                         |
| Register / Roll No. | 192424060                             |
| Role in the Team    | UI and Interactive Visualization Lead |

#### C. Task-wise Individual Contribution

*List every task/module you personally worked on. Add rows if needed. Do not describe work done by teammates.*

| S.No. | Task / Module Handled        | Description of Work Done                                                              | Tools / Techniques Used | Approx. Hours |
|-------|------------------------------|---------------------------------------------------------------------------------------|-------------------------|---------------|
| 1     | Website Structure            | Created and organized the main sections of the assignment website.                    | HTML                    | 4             |
| 2     | User Interface Development   | Designed the layout for algorithm information, controls, results and solver sections. | HTML, CSS               | 5             |
| 3     | Graph Visualization          | Created the visual representation of courses and conflicts using nodes and edges.     | SVG, HTML, CSS          | 5             |
| 4     | Interactive Solver Interface | Developed the interface for selecting colours and running the solver.                 | JavaScript, DOM         | 5             |
| 5     | Results Display              | Worked on displaying colouring assignments and solver results clearly.                | JavaScript, CSS, SVG    | 4             |

#### D. Team Contribution Percentage Summary

To be filled identically by every team member so contributions are consistent and sum to 100%. Disagreements should be discussed and resolved before submission.

| Team Member Name        | Register No.        | % Contribution        | Signature        |
|-------------------------|---------------------|-----------------------|------------------|
| <b>Team Member Name</b> | <b>Register No.</b> | <b>% Contribution</b> | <b>Signature</b> |
| Haresh Kumar N L        | 192425009           | 35%                   | Haresh Kumar     |
| Mohan Kumar J           | 192424060           | 35%                   | Mohan Kumar      |
| Nirmal Kasi             | 192424248           | 30%                   | Nirmal           |
| <b>Total</b>            |                     |                       | <b>100%</b>      |

## E. Course Outcomes / Skills Applied

| Course Outcome(s) Addressed by MY Work                     | How I Applied It                                                                                                                                                                                                                                                                                                                 |
|------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CO1 – Understanding of Algorithmic Problem Solving</b>  | I applied the concepts discussed in the course by understanding how the Graph Colouring solution can be represented and demonstrated through an interactive application. I worked on presenting the relationship between courses, conflicts, colours and timetable slots in a visual and understandable manner.                  |
| <b>CO2 – Algorithm Design and Analysis</b>                 | I supported the presentation of the algorithmic solution by creating an interface through which the Backtracking and pruning-based approaches could be demonstrated. The website structure was designed to clearly separate the problem formulation, algorithm explanation, complexity discussion, comparison and solver output. |
| <b>CO3 – Application of Data Structures and Algorithms</b> | I applied the concepts of graph representation and Graph Colouring while developing the visual graph. The courses were represented as nodes and the conflicts between courses were represented as edges. This allowed the theoretical graph structure used by the algorithm to be displayed visually on the website.             |
| <b>CO4 – Implementation of Algorithms</b>                  | I contributed to integrating the implemented algorithm with the interactive website. I worked with JavaScript-based controls and dynamic webpage elements so that users could interact with the solver and view the colouring results. This helped convert the theoretical algorithm into a demonstrable software application.   |

## F. Challenges Faced and How I Resolved Them

Describe any technical or teamwork challenges you personally faced, and the steps you took to resolve them.

One of the main challenges I personally faced was designing the website so that a technical algorithm could be presented in a simple and understandable manner. The project involves concepts such as Graph Colouring, Backtracking, pruning, complexity analysis and performance comparison. If all of this information were placed together without proper organization, the website could become difficult to follow. I resolved this by dividing the interface into clearly separated sections for the problem, algorithm methodology, comparison, complexity, results and interactive solver.

Another significant challenge was the **visual representation of the conflict graph**. The courses had to be displayed as individual nodes while the conflicts between them had to be represented using edges. The positioning of the edges had to correspond accurately with the node positions. During development, I encountered alignment issues where the graph edges did not visually match the positions of the course nodes. I resolved this by using a common SVG coordinate system for both nodes and edges, ensuring that the graphical elements were positioned consistently.

I also faced challenges while working on the **interactive solver interface**. The interface needed to allow the user to interact with the available controls and then display the appropriate colouring result. This required coordination between the webpage elements and JavaScript functionality. I worked on connecting the controls with the solver output and ensuring that the displayed information was updated when the user interacted with the application.

Another challenge was making the graph visualization readable without making it unnecessarily large. If nodes were too large or placed too close together, the graph became cluttered and the edges were difficult to follow. I refined the node sizes, spacing and overall positioning so that the graph could be understood more easily.

I also worked on improving the overall visual consistency of the website. Different sections needed consistent spacing, typography, borders and panel structures. I refined the CSS styling to make the website more organized and professional while keeping the important technical information visible.

Responsive behaviour was another consideration. The website needed to remain usable when viewed on different screen sizes. I therefore adjusted the layout and graphical elements so that the content did not become excessively compressed or difficult to read on smaller displays.

Finally, I performed repeated browser-based testing after integrating the different components. This helped me identify issues with graph positioning, visual spacing and interactive elements. I resolved these issues through incremental changes and testing rather than changing multiple components simultaneously, which made it easier to identify the source of each problem.

## G. Individual Learning Outcome / Reflection

**What did YOU personally learn from completing your part of this assignment? How did it build on classroom concepts?**

Through my contribution to this assignment, I gained practical experience in converting an algorithmic problem into an **interactive web-based application**. In classroom learning, algorithms are often represented using pseudocode, flowcharts, mathematical notation and theoretical examples. Working on the website helped me understand how these concepts can be communicated to a user through an actual interface.

One of the most important things I learned was how **Graph Colouring can be represented visually**. Instead of only describing courses and their conflicts using text, I worked on displaying each course as a graph node and each conflict as an edge. This helped me understand the relationship between the abstract graph structure and its visual representation.

I also developed stronger practical skills in **HTML, CSS and JavaScript**. I learned how HTML can be used to organize technical content, how CSS can control layout and presentation, and how JavaScript can provide interaction and dynamic updates. Combining these technologies showed me how different layers of a software application work together.

Working with SVG also provided useful experience in graphical programming. I learned that visual elements such as nodes and edges must be positioned using consistent coordinates. Small inconsistencies in positioning can make an otherwise correct graph look incorrect. Solving these alignment problems improved my debugging and attention-to-detail skills.

The interactive solver also helped me understand the importance of **user interaction in algorithm demonstration**. Instead of simply showing a final answer, an interactive interface allows users to change inputs and observe the resulting behaviour. This makes the algorithm easier to explore and provides a more practical demonstration of the concepts learned in the course.

Another major learning outcome was understanding that **software usability is separate from algorithm correctness**. An algorithm may produce the correct solution, but if its output is difficult to understand, the application is less effective. I therefore learned to consider visual hierarchy, spacing, readability and feedback while developing the interface.

The project also strengthened my debugging skills. When the graph visualization did not appear correctly, I had to examine the relationship between the SVG coordinates, node positions and edge paths. This taught me to approach interface problems systematically instead of making random changes and hoping the browser suddenly develops sympathy.

Overall, this assignment helped me connect classroom concepts in algorithms and graph theory with practical programming and visualization. I gained experience not only in developing a functional interface but also in presenting an algorithm in a way that makes its working and results easier for others to understand.

## H. Declaration

I declare that the contribution described above accurately represents the work I personally performed for this team assignment, and that I have not misrepresented the work of any other team member as my own.

| Student Signature | Date       |
|-------------------|------------|
| Mohan Kumar J     | 09/02/2026 |